Student's Index Number ……

THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITIES

ARUSHA REGIONAL MOCK EXAMINATION

FORM TWO

032 **CHEMISTRY**

TIME: 2:30HRS **MAY 2026**

INSTRUCTIONS

1. This paper consists of sections A, B and C
2. Answer questions in both sections on the spaces provided
3. All writing must be in black / blue except diagrams which must be in pencil.
4. Cellular phones and any authorized materials are not allowed in the examination room
5. The following constant may be used.
   Atomic masses: H = 1, C = 12, N = 14, O = 16

FOR ASSESOR USE ONLY		
QUESTION NUMBER	**SCORE**	**ASSESORS INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

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SECTION A (15 Marks)

Answer All questions in this section

1. For items (i) to (x), choose the most correct answer from the among given alternatives and write its letter in the box provided below.

(i) Some experiments require powderly chemicals which apparatus will you use to make them available from the given granules
A. Pestle and filter funnel
B. Round bottomed flask and trough
C. Motor and Pestle
D. Crucible and Lid

(ii) The chief cook added salt to the cup of water.The resulting solution can be termed as: -
A. Homogenous mixture
B. Binary compound
C. Heterogenous mixture
D. Uniform suspensions

(iii) What are nucleons?
A. Neutrons and electrons
B. Neutrons and Protons
C. Electrons and protons
D. Protons, neutrons and electrons

(iv) Which property is the basis for use of carbon -14 in archaeology for dating ancient objects?
A. Electronic configuration
B. Radioactive decay
C. Atomic mass
D. Relative atomic mass

(v) The electronic arrangement of an element is 2:3. This element is found in: -
A. Group III, period 2
B. Group VI, period 2
C. Group VII, period 3
D. Period 3, group II

(vi) Group I elements burns in air to form.
A. Metal oxides
B. Non – metal oxides
C. Hydroxides
D. Carbon dioxide

(vii) What happens during chemical bonding?
A. Non – metals gains protons
B. Metals looses their outermost shell electrons
C. Metals gains electrons in their outermost shell
D. Non – metals losses electrons of their outermost shells

(viii) A certain metals hydroxide reacts with hydrochloric acid to form salt and water only. What type of reaction is this
A. Precipitation
B. Displacement
C. Combination
D. Neutralization

(ix) Changing in states of matter involved when drying wet clothes is: -
A. Liquid to gas
B. Solid to liquid
C. Gas to liquid
D. Solid to gas

(x) A track of Kerosene gets an accident, and unfortunately fire breaks out. People wanted to extinguish it using water but they were told not to use water because water would spread the fire. This is because: -
A. Kerosine floats over water
B. Water floats over kerosine

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C. Water is less denser that kerosine
D. Water is neutral to fire.

Answer

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the items in LIST A with the correct response from LIST B by writing the letter of the correct response besides the item number in the box provided .

LIST A	LIST B
(i) Used for holding, heating and estimating the volume of liquids (ii) Used for measuring the mass of chemicals in the laboratory (iii) Used as a source of heat in the laboratory (iv) Used for holding substances that are being weighed or observed (v) Used for sucking in and measuring the volume of liquids or gases	A. Thistle funnel B. Watch glass C. Measuring cylinder D. Test tube E. Beaker F. Conical flask G. Bunsen burner H. Electronic balance

Answer

LIST A	**i**	**ii**	**iii**	**iv**	**v**
LIST B					

SECTION B (70 Marks)

3. (a) You are provided with the hypothetical element M and N, with the atomic number 12 and 17 respectively
   (i) Write the chemical formula of the compound formed by the combination form by M and N.
   ……
   (ii) Mention the type of bond formed between element M and N …….
   (iii) State any three properties of compound formed in (i) above
   …….
   …….
   …….
   (b) Three containers, A , B and C which contain concentrated Sulphuric acid, rat poison and petrol respectively were on the table without warning signs. By using chemistry knowledge identify and draw the warning signs which would appear on container A, B and C.
   (i) Warning signs container A

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(ii) Warning signs on container B

(iii)Warning sign on container C

4. (a)Medical doctors uses x -rays to view patients bodies. These rays interacts with atoms in the patients bones.
   (i) Which sub atomic particles is responsible for this interaction?.....
   (ii) Why do you think this particle is important in medical imaging? Gie two reasons

   ……
   ……

   (b) Suggest the best method used to separate the following mixtures.

   (i) Iodine and sand ……..
   (ii) Kerosine and sand ……..
   (iii)Oil from seeds……
   (iv)Alcohol and water ……..
   (v) Chlorophyl from leaves ……
   (vi)Sugar and water……..

5. (a) Define the following terms
   (i) Electronegativity ……...……...

   (ii) Ionization energy ……

   (iii)

   (b) Study the following part of the periodic table and then answer the questions that follows:

I							VIII
	II	III	IV	V	VI	VII	

Place the elements having the atomic number 1,9,10,14,19 and 20 in the periodic table by using letters A,B,C,D respectively.

- From the periodic table identify the element which: -

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(i) Belong to alkaline earth metal bock …….
(ii) Is a noble gas ……
(iii) Burns in oxygen to form water……
(iv) Has highest electronegativity ……..
(v) Has highest atomic size……..
(vi) Has a valency of -4 …….

6. (a) An isotope f element x has the electronic configuration of 2:8:5 and mass number of 31.
   (i) Classify the group and period of element x ……
   (ii) How many electrons does it have? ……
   (iii)How many nuetrons does it have?.....
   (iv)Is element x a metal o non – metal? Give reasons for your answer ……..
   ……
   ……
   (v) Write the nuclide notation for element x…….

   (b) Neon has three isotopes ^{20}Ne (90.5%) and ^{21}Ne (0.3%) and ^{22}Ne (9.2%) calculate its relative atomic mass.

7. (a) Differentiate oxidation and reduction in terms of electrons transfer.
   ……
   ……
   (b) During debate session, form three students were given a motion which says that Molecular equations are better than word equation . Support the motion by giving four points.
   (i) ……..
   (ii) ……..
   (iii)……..
   (iv)……..

   (c) Identify the type of reaction in each of the following chemical equations.

   (i) $CuCO_{3(s)} \longrightarrow CuO_{(s)} + CO_{2(g)}$
   (ii) $CaO_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_{2(aq)}$
   (iii) $Zn_{(s)} + CuSO_4 \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$
   (iv) $Zn(NO_3)_{2(aq)} + NaOH_{(aq)} \longrightarrow Zn(OH)_{2(s)} + NaNO_{3(aq)}$

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8. (a) We encounter many events / procedures in our daily life that are good examples of chemical changes. Give three examples of these changes
   (i) ……..
   (ii) ……..
   (iii)…….

(b) List down four (4) properties that clearly justify that water is a compound

   (i) ……..
   (ii) ……..
   (iii)…….
   (iv)……

(c) Acid are found naturally I many substances. Name the acid found in the following substances.

   (i) Sour milk ……..
   (ii) Bee sting ……
   (iii)Spinach ……..
   (iv)Citrus fruit……
   (v) Nettle plant ……
   (vi)Grapes……

9. (a) Identify three common accidents which can occur in a chemistry laboratory.
   (i) . ……..
   (ii) ……..
   (iii)…….

(b) Advice a head of the school who wants to build chemistry laboratory on the three qualities of a good laboratory.
   (i) . ……..
   (ii) ……..
   (iii)…….

(c) Form one students were told by their teacher to prepare themselves for practical activity list an four laboratory rules they should observe during practical activity.
   (i) . ……..
   (ii) ……..
   (iii)…….
   (iv). ……..

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SECTION C (15 Marks)

10. (a) Define the following terms

(i)Empirical formula

……

……

(ii) Molecular formula …….

……

(b) Write the chemical formula for the following compounds

(i)Sodium oxide …….

(ii) Calcium nitrate……..

(iii) Ammonium chloride…….

(iv) Potassium cyanide …….

(c) A certain gaseous compound contains 30.4% of Nitrogen and 69.6% of oxygen by mass. If the vapour density of the compound is 46, calculate .

(i) Its empirical formula

(ii) Its molecular formula

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(iii) Give he name for the compound.

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